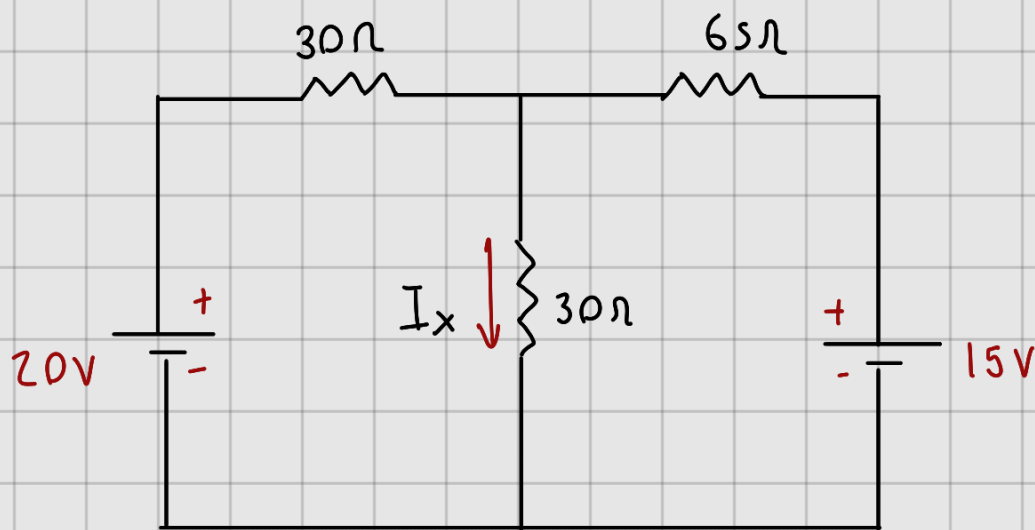
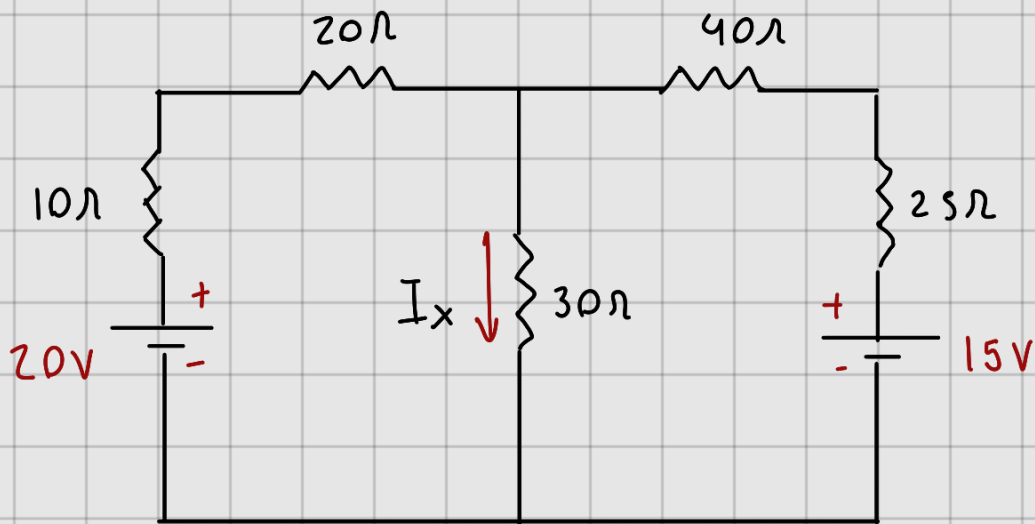
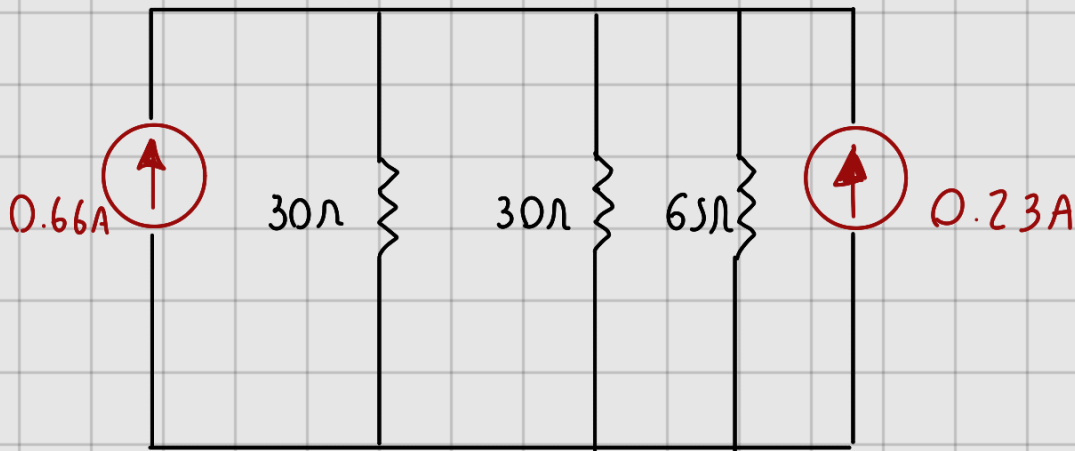




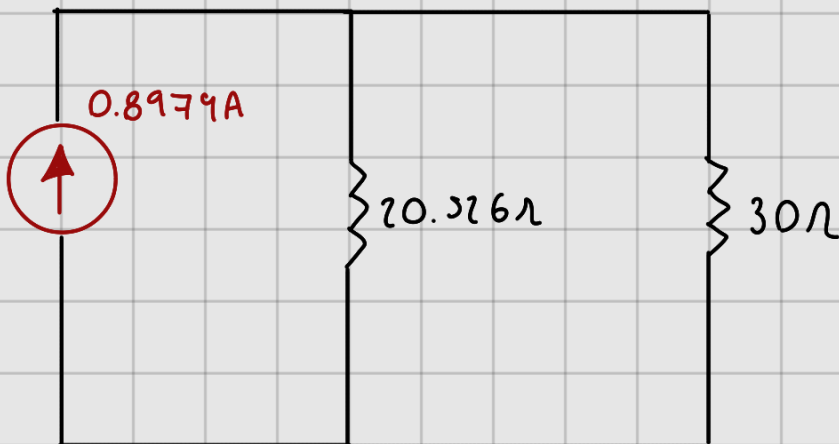
$$I_s = \frac{V_s}{R} = \frac{20V}{30\Omega} = 0.66A$$

$$I_s = \frac{V_s}{R} = \frac{15V}{65\Omega} = 0.23A$$

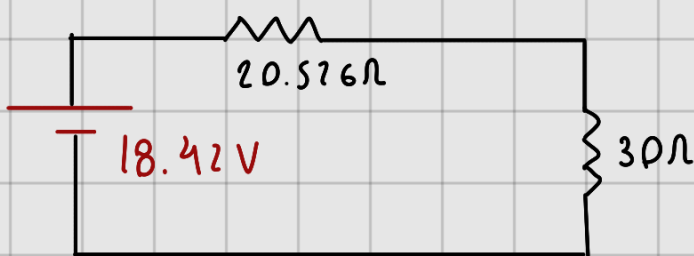


$$I_T = 0.66A + 0.23A = 0.8974A$$

$$R_A = \frac{65\Omega * 30\Omega}{65\Omega + 30\Omega} = \frac{1950}{95} = 20.526\Omega$$



$$V_s = I_s R = 0.8974 (20.526\Omega) = 18.42V$$



$$R_T = 50.526\Omega$$

$$V_T = 18.42V$$

$$I_T = \frac{V_T}{R_T} = \frac{18.42V}{50.526\Omega} = 0.364A$$

$$\underline{I_x = 0.364A}$$